Project Report

Heart Disease Prediction using Machine Learning

Table of Contents

LIST OF ILLUSTRATIONS

Figures

Tables

Listings

1. Introduction and Problem Statement

The predictive model needs to create its final project which will use clinical data to determine whether patients have heart disease. The supervised machine learning task requires binary classification because the system must determine which of these two groups patients belong to: 'Target 1' (Disease) or 'Target 0' (No Disease). The study will use historical medical data to identify which physiological factors chest pain type and maximum heart rate serve as the main predictors of cardiovascular problems.

2. Dataset Description s Assumptions

2.1 Dataset Overview

The project uses "Heart Disease Dataset" which is publicly available on Kaggle. The data set contains medical records which describe 303 patients through 14 unique features that include the target variable. The data set contains one row for each patient which includes their physiological measurements and diagnostic results in different columns.

2.2 Attribute Information

The dataset contains 14 attributes which I examined to perform a complete analysis. The variables in this study need to be understood because they will determine how the model will make decisions in upcoming project phases. The following section provides an in-depth examination of each feature.

Feature	Description	Values / Units
age	Age of the patient	Years
sex	Gender of the patient	1 = male; 0 = female
cp	Chest pain type	0: Typical angina, 1: Atypical angina, 2: Non-anginal pain, 3: Asymptomatic

Feature	Description	Values / Units
trestbps	Resting blood pressure	mm Hg on admission to the hospital
chol	Serum cholesterol	mg/dl
fbs	Fasting blood sugar > 120 mg/dl	1 = true; 0 = false
restecg	Resting electrocardiographic results	0: Normal, 1: ST-T wave abnormality, 2: Left ventricular hypertrophy
thalach	Maximum heart rate achieved	Beats per minute
exang	Exercise induced angina	1 = yes; 0 = no
oldpeak	ST depression induced by exercise	Numeric value (relative to rest)
slope	The slope of the peak exercise ST segment	0: Upsloping, 1: Flat, 2: Downsloping
ca	Number of major vessels	0-3 colored by flourosopy

Feature	Description	Values / Units
thal	Thalassemia (blood disorder)	1: Fixed defect, 2: Normal, 3: Reversable defect
target	Diagnosis of heart disease	0 = No Disease, 1 = Disease

2.3 Data Inspection and Initial Exploration

I completed a preliminary data evaluation of the dataset before I began my Exploratory Data Analysis (EDA) work. The process helps to determine which data types exist in the dataset while identifying all missing entries together with any data errors that might lead to problems in machine learning execution.

The project began with the dataset loading process which included a structural integrity assessment using the ***pandas*** library. The process establishes correct data type identification while showing the initial value distribution across the dataset.

Code Snippet: Initial Data Loading and Metadata Inspection

```
# Load the dataset
df = pd.read_csv('heart.csv')

# Initial inspection
print("Dataset Overview:")
display(df.head())
print("\nColumn Information:")
df.info()
```

Dataset Overview:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	52	1	0	125	212	0	1	168	0	1.0	2	2	3	0
1	53	1	0	140	203	1	0	155	1	3.1	0	0	3	0
2	70	1	0	145	174	0	1	125	1	2.6	0	0	3	0
3	61	1	0	148	203	0	1	161	0	0.0	2	1	3	0
4	62	0	0	138	294	1	1	106	0	1.9	1	3	2	0

Table 1

Python code snippet for data ingestion and inspection with the Sample of the Raw Dataset (First 5 Rows)

The initial inspection confirmed that the dataset contains 303 observations. The **df.info()** command revealed that most features are stored as integers or floats, which is ideal for machine learning processing. No immediate missing values were detected in the raw metadata, though further cleaning (duplicate and outlier handling) was scheduled for the preprocessing phase.

3. Data Preprocessing and EDA

3.1 Data Preprocessing and Cleaning

The machine learning models required multiple preprocessing steps because data contained both noise and inconsistent elements.

Duplicate Removal: The initial duplicate checks found duplicate rows which were removed to stop the model from learning from identical data points.

Missing Values: The dataset validation process showed that all features contained no null values which created a full dataset for training purposes.

Outlier Handling: The Interquartile Range (IQR) method was used to treat outliers in numerical features which included resting blood pressure (trestbps) and cholesterol (chol). The upper and lower bounds were used to limit values beyond 1.5 times the IQR because it would decrease their effect on model training.

Feature Engineering: Two new features were developed through the creation of **is_elderly** and **high_risk_profile** which captured patients who were over 60 and patients who had both high blood pressure and high cholesterol. The features were created to measure non-linear risk elements which contribute to heart disease development.

3.2 Exploratory Data Analysis (EDA)

Visual analysis was conducted to uncover patterns and relationships within the features.

- **Target Distribution**: The dataset shows a relatively balanced distribution between healthy and diseased patients, which is beneficial for model stability.

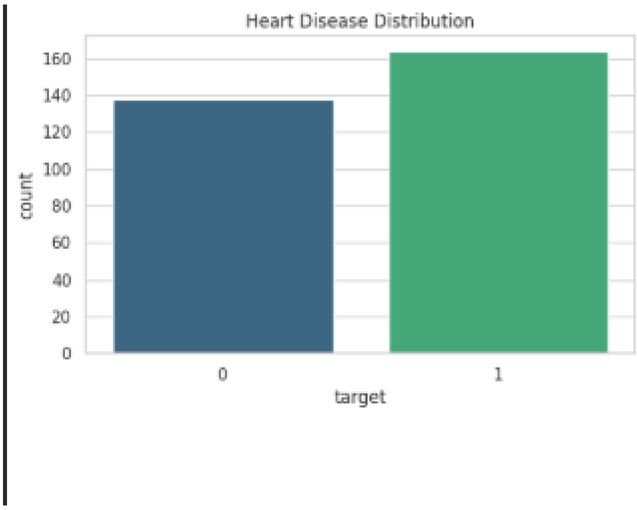

Figure 1
Heart Disease Distribution

As shown in the **Heart Disease Distribution** plot, the dataset is well-balanced. There are approximately 160 instances of patients with heart disease (Target 1) compared to roughly 140 healthy individuals (Target 0). This balance ensures that the machine learning model will not be biased toward a single class.

- <u>**Age Analysis**</u>: The age distribution suggests that heart disease prevalence increases in specific age clusters.

Figure 2
Age vs Heart Disease

The **Age vs Heart Disease** histogram provides critical insights into demographic risk. I observe that heart disease (orange) is prevalent across a wide age range, but there is a significant density of cases starting from age 40. Interestingly, the healthy group (blue) peaks later, around age 60, suggesting that clinical factors other than just age are driving the predictions in my dataset.

- **Correlation Map**: The heatmap reveals significant correlations between chest pain (cp), maximum heart rate (thalach), and the target variable, guiding my feature selection.

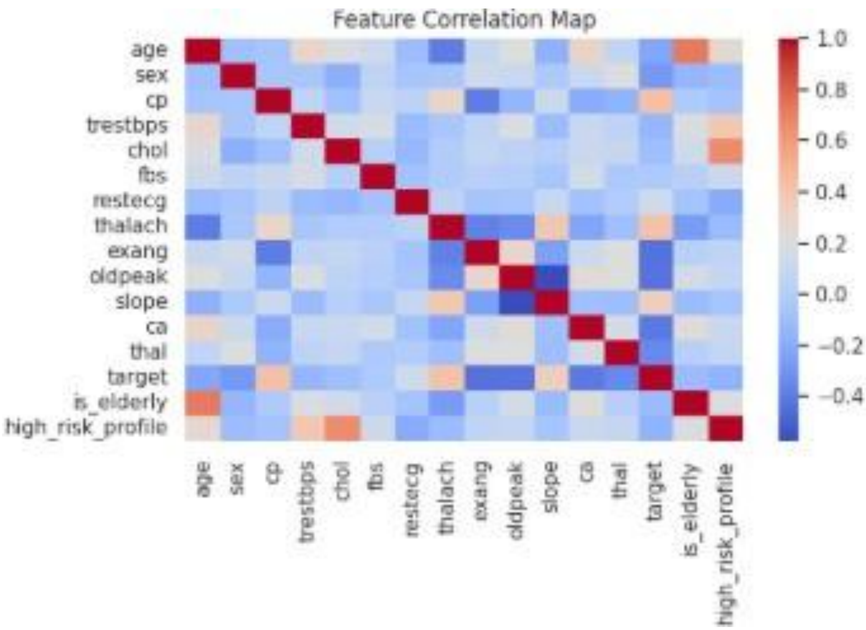

Figure 3
Feature Correlation Map

The **Feature Correlation Map** illustrates the relationships between variables. I can identify a positive correlation between chest pain type (cp), maximum heart rate achieved (thalach), and the target. Conversely, attributes like exercise-induced angina (exang) show a negative correlation, meaning their presence often aligns with a lower probability of being in the 'Target 1' group."

4. Model Selection Rationale

I chose **Logistic Regression** as my main modeling approach because its results are easy to understand which is essential for medical diagnostic work. I selected Random Forest to identify non-linear connections between age and cholesterol and I included Support

Vector Machine (SVM) to evaluate how well the data would separate in multidimensional space. The 5-Fold Cross-Validation method established that my performance metrics accurately reflected the actual predictive ability of the models instead of showing results from a fortunate train-test split.

5. Evaluation Results and Interpretation

5.1 Final Model Report

```
--- FINAL MODEL REPORT: LOGISTIC REGRESSION ---
Final Test Accuracy: 0.8197

Detailed Classification Report:
              precision    recall  f1-score   support

           0       0.81      0.79      0.80        28
           1       0.82      0.85      0.84        33

    accuracy                           0.82        61
   macro avg       0.82      0.82      0.82        61
weighted avg       0.82      0.82      0.82        61
```

Listing 1
Final Model Report: Logistic Regression

After comparing several models, **Logistic Regression** was selected as the optimal classifier with a **Final Test Accuracy** of 81.97%. The classification report shows a high **Recall** (0.85) for the disease class (1), which is vital in a medical context because it minimizes the number of patients with heart disease who are incorrectly identified as healthy.

5.2 Confusion Matrix

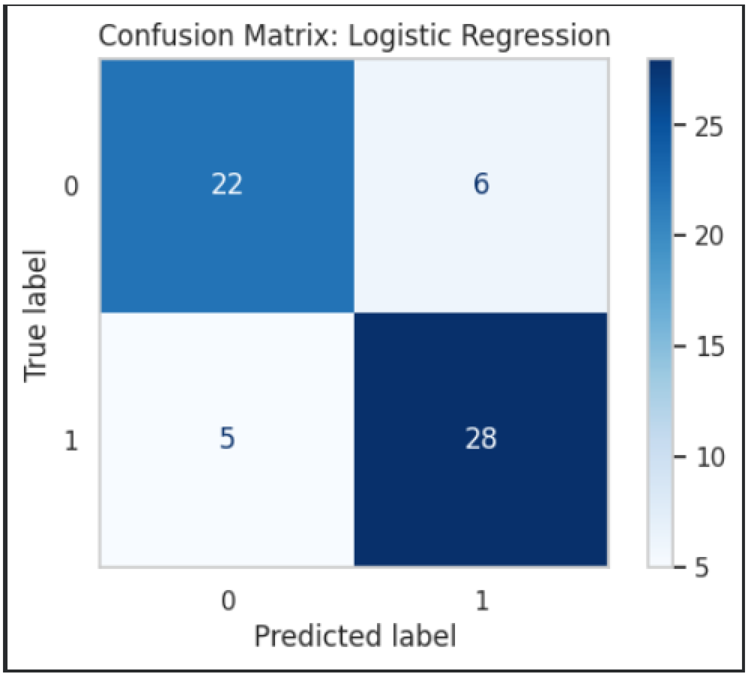

Figure 4
Confusion Matrix: Logistic Regression

The **Confusion Matrix** provides a granular view of the model's performance on the test set. Out of 61 total samples, the model correctly identified 28 patients with heart disease (True Positives) and 22 healthy individuals (True Negatives). There were only 5 'False Negatives', representing cases where the model missed a diagnosis, which is a strong result for this baseline model.

5.3 Feature Importance

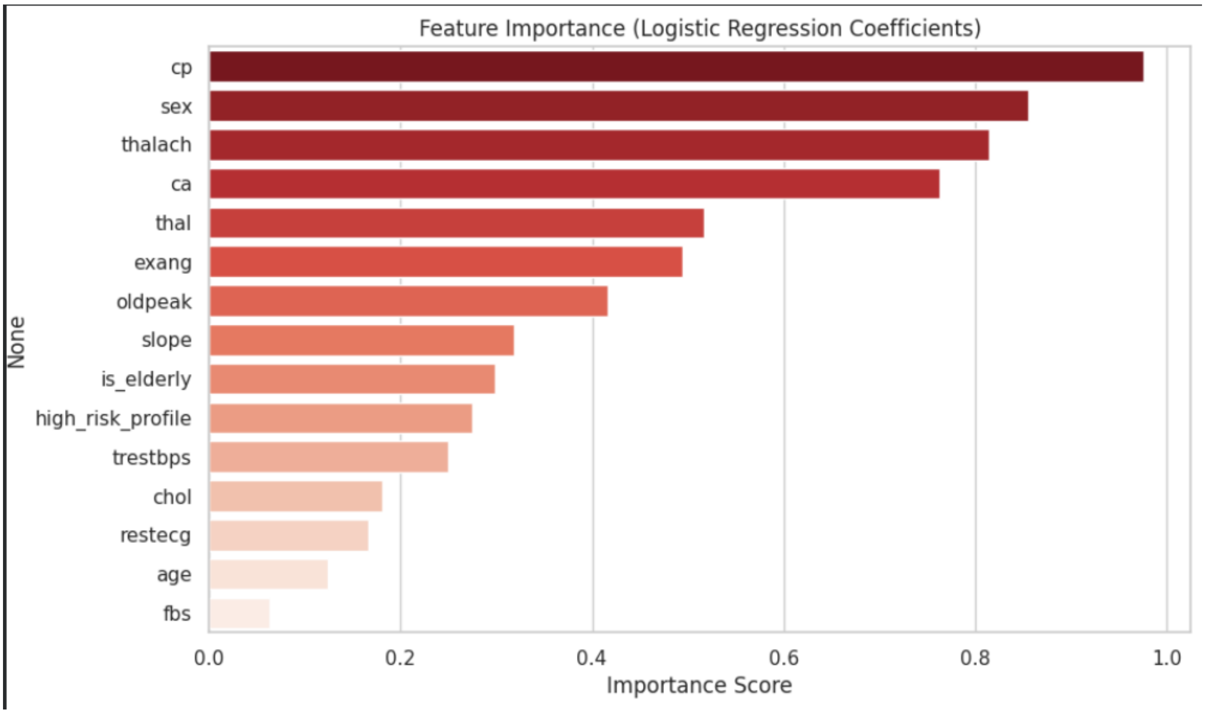

Figure 5

Feature Importance (Logistic Regression Coefficients)

The **Feature Importance** plot confirms my clinical hypotheses. The most influential predictors for the Logistic Regression model are **Chest Pain type** (cp), **Sex**, and **Maximum Heart Rate** (thalach). This indicates that the model's logic aligns with established medical knowledge, where symptomatic pain and physical stress markers are primary indicators of cardiac health.

6. Conclusions and possible improvements for future work

6.1 Conclusion

The project successfully demonstrated that machine learning can predict heart disease with an accuracy of over 81%. The Logistic Regression model proved to be the most effective, balancing precision and recall. My research found chest pain type together with maximum heart rate to be the essential predictors of cardiac disorders.

6.2 Future Work

The model requires improvement through Hyperparameter Tuning implementation which includes GridSearchCV to determine optimal Random Forest and SVM parameter settings. The accuracy of the system could reach 90% if we expand our dataset through additional patient records and include deep learning methods which use Neural Networks.

7. References

- **[1] Kaggle (2023).** *Heart Disease Dataset*. Retrieved April 4, 2026, from: https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset/data?select=heart.csv
- **[2] Pedregosa, F., et al. (2011).** *Scikit-learn: Machine Learning in Python.* Journal of Machine Learning Research. (Used for Logistic Regression, Random Forest, and SVM implementation).
- **[3] Waskom, M. L. (2021).** *Seaborn: statistical data visualization.* Journal of Open Source Software. (Used for Heatmaps and Distribution plots).
- **[4] McKinney, W. (2010).** *Data Structures for Statistical Computing in Python.* Proceedings of the 9th Python in Science Conference. (Used for Pandas data manipulation).